When Does Online Estimation of Persistent Capacity Improve Dynamic Crowd Guidance? A Mesoscopic Queue-Network Study

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***Abstract*—Dynamic crowd guidance may improve through reactive balancing, predictive lookahead, or online information. We quantify the incremental value of estimating persistent but initially unknown capacities in a mesoscopic queue-network benchmark where both controllers already observe current queues. The primary comparison updates capacity-scenario centers within a shared rolling-horizon design, retaining the same pre-clipping log-spread, clipping rule, and random-number coupling. Across 200 paired validation runs, conditioning reduced mean clearance time by 15.65 s (2.17%; conditioned-minus-unconditioned 95% paired-bootstrap interval $[-20.75, -10.80]$ s). The direct conditioned-minus-reactive effect was -3.00 s ($[-7.50, +1.35]$ s), leaving its clearance advantage over reactive guidance unresolved. Estimator diagnostics distinguish well-observed exits from sparsely observed paths. Independent same-state counterfactuals examine candidate scoring, and a three-policy runtime comparison measures computational cost. The evidence supports a modest within-controller benefit under the specified synthetic assumptions; it does not establish general superiority of predictive guidance or operational crowd-safety performance.**

***Index Terms*—crowd egress, dynamic guidance, information value, queue network, model predictive control**

I. Introduction

Dynamic crowd guidance is often justified by live sensing, yet improved performance may instead arise from reactive balancing, predictive lookahead, or a different optimization procedure. This paper asks a narrower question: *when does online estimation of effective capacity add value to an otherwise matched dynamic controller?* The primary experiments concern persistent heterogeneity. Path and exit capacity multipliers are sampled once at the start of a run and remain fixed; saturated-throughput observations update their estimates online. Abrupt capacity changes are outside the primary claim.

We separate static routing, deterministic reactive balancing, unconditioned rolling-horizon mean control, and throughput-conditioned mean control. A non-deployable oracle supplies a diagnostic benchmark. For run-level clearance time T, the primary estimand is

$$\Delta T_{\text{sense}} = \mathbb{E}[T_{\text{cond mean}} - T_{\text{uncond mean}}], \tag{1}$$

so negative values favor conditioning. Both rolling-horizon controllers observe the current queue state; Eq. (1) measures the additional value of capacity estimates given that state information. The intervention changes the pre-clipping prediction center under a common log-spread and clipping rule.

The contributions are: (i) a matched capacity-center ablation with common random numbers; (ii) paired validation and direct diagnostics of estimation error and link observability; and (iii) same-state candidate-selection diagnostics, computational-cost measurements, and operating-region studies that delimit the benefit. We distinguish this within-controller estimand from the secondary system-level comparison with reactive guidance. Code, configurations, frozen CSV tables, and figure manifests are available in `event-egress-python`.

II. Related Work

Crowd evacuation and guidance models include microscopic agents, cellular automata, social-force simulation, network flow, digital twins, reinforcement learning, and optimization [1], [2], [3], [4]. Route-choice research emphasizes bounded rationality, visual information, social factors, and communication [5], [6], [7]. Robust and uncertainty-aware pedestrian modeling is also established [8], [9].

Adaptive guidance, simulation-based optimization, and compliance sensitivity are established in CellEVAC and related cell-based methods [10], [11]. Other approaches use smart-building graphs [12], dynamic signs [13], or pedestrian-flow MPC. Lopez-Carmona and Paricio Garcia compare linear and nonlinear predictors with discrete speed instructions [14]. Their MISO study compares models with measured pedestrian inflow against SISO models that treat inflow as an unmeasured disturbance, finding prediction benefits that depend on the horizon [15]. Additional measurements in crowd MPC are therefore not a new research topic.

Our specific comparison concerns the closed-loop clearance effect of updating persistent-capacity forecast centers for route-share guidance while retaining a common controller configuration, pre-clipping spread, clipping rule, and random draws. It complements predictor-accuracy and algorithm comparisons. Learning is passive: the smoother uses naturally occurring saturated flows, without optimizing actions for information acquisition as in dual adaptive MPC [16]. Empirical calibration remains a separate challenge [17], [18].

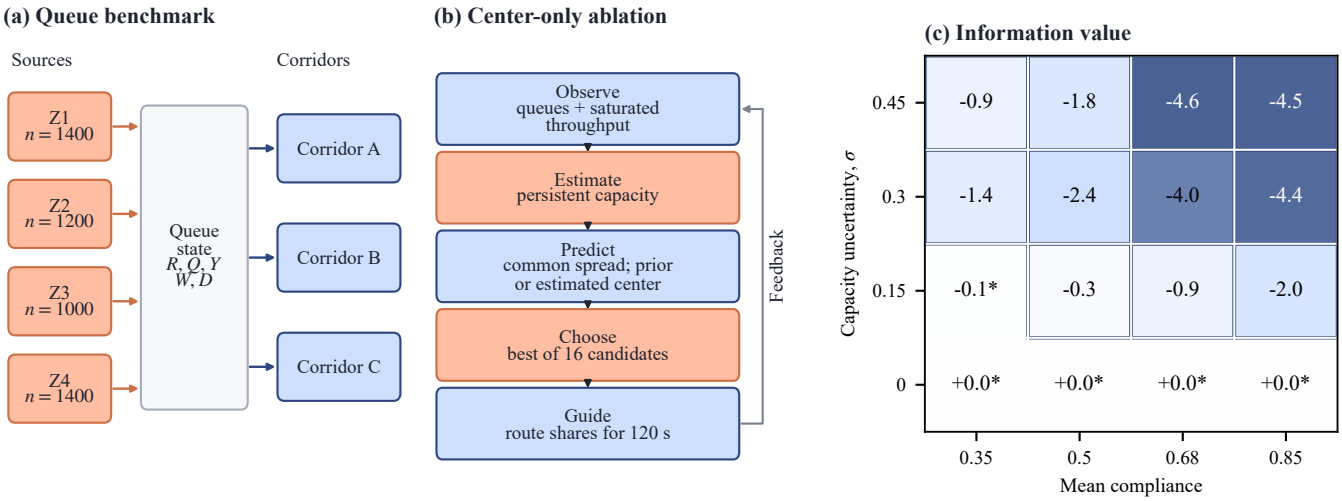


Fig. 1. Benchmark and information value. (a) Base queue network. (b) Matched feedback controller. (c) Conditioned-minus-unconditioned mean clearance effect (%): blue is negative and orange positive; negative favors conditioning. Stars mark 95% paired-bootstrap intervals containing zero, including $[0, 0]$; blue cell borders mark intervals entirely below zero.

III. Model and Online Information

Let $\mathcal{Z}$ denote source zones and $\mathcal{E}$ destination corridors. The conserved state contains unreleased populations R_z, path queues Q_{ze}, travel-delay cells Y^k_{ze}, exit queues W_e, and cumulative discharge D. Recommendations $u_{ze}(t)$ mix with one-hot habitual routing h_{ze} through compliance ρ_z:

$$p_{ze}(t) = \rho_z u_{ze}(t) + (1-\rho_z)h_{ze}, \qquad \sum_e p_{ze}(t) = 1. \quad (2)$$

For nominal path capacity c_{ze}, exit capacity C_e, and fixed multipliers ξ_{ze}, ζ_e,

$$\widetilde{Q}_{ze}(t) = Q_{ze}(t) + b_z(t)p_{ze}(t), \quad (3)$$

$$x_{ze}(t) = \min\{\widetilde{Q}_{ze}(t), c_{ze}\xi_{ze}\Delta t\}, \quad (4)$$

$$y_e(t) = \min\{W'_e(t), C_e\zeta_e\Delta t\}. \quad (5)$$

Here $b_z(t)$ is released population during the step and $W'_e(t)$ includes arrivals from existing travel cells. Existing delay cells shift before newly admitted path flow enters the final delay cell, so new flow cannot reach an exit in the same step. Queue and discharge updates conserve total population; clearance occurs when active mass falls below 10^{-8} persons.

For $m_j \in \{\xi_{ze}, \zeta_e\}$,

$$\log \widetilde{m}_j \sim \mathcal{N}(-\sigma^2/2, \sigma^2), \quad m_j = \mathrm{clip}_{[0.45,1.35]}(\widetilde{m}_j), \quad (6)$$

$$\widetilde{\rho}_z \sim \mathrm{Beta}(\mu\kappa, (1-\mu)\kappa), \quad \rho_z = \mathrm{clip}_{[0.02,0.98]}(\widetilde{\rho}_z), \quad (7)$$

with default $\mu = 0.68$ and $\kappa = 18$. All queue states are observed. A capacity estimate updates only when demand saturates the corresponding path or exit:

$$\widehat{m}(t+1) = (1-a)\widehat{m}(t) + a\frac{x(t)}{c\Delta t}, \qquad a = 0.3. \quad (8)$$

Saturation is classified using demand and the true step capacity in the simulator. Under this favorable, noise-free observation model, a saturated throughput identifies the true multiplier before smoothing. After k saturated observations,

$$\widehat{m}_k - m = (1-a)^k(\widehat{m}_0 - m). \quad (9)$$

An unsaturated observation instead supplies a lower bound on capacity: serving all demand D implies $cm\Delta t \geq D$. It does not point-identify m, and the present estimator leaves its estimate unchanged. Estimator accuracy must therefore be interpreted jointly with saturation coverage; accurate bottleneck estimates need not imply identification of every link.

IV. Guidance and Evaluation

The reactive baseline assigns route shares using free-flow time and current path/exit workload. Rolling-horizon policies use the same 16-candidate construction: habitual and reactive recommendations, mixtures of the reactive recommendation with equal shares, capacity–travel-time shares or the previous recommendation, and Dirichlet perturbations. Each candidate recommendation is held fixed during its eight-scenario, 300-s rollout. For terminally corrected clearance estimate $\widehat{T}_s$, mean person time $\bar{A}_s$, and peak queue-density proxy d_s, the score is

$$L_s(u) = \widehat{T}_s(u) + w_A\bar{A}_s(u) + w_d[d_s(u) - d_0]^2_+, \quad (10)$$

$$J(u) = S^{-1}\sum_s L_s(u) + \lambda\,\mathrm{CVaR}_{0.8}(L(u)) + w_s\,\mathrm{mean}_{z,e}\,|u_{ze} - u^{\mathrm{prev}}_{ze}|. \quad (11)$$

Clearance and mean person time are measured in minutes. The default weights are $w_A = 0.25$, $w_d = 6$ and $w_s = 0.08$, with density threshold $d_0 = 3.5$; the primary policies set $\lambda = 0$. The clearance surrogate adds terminal path work, exit work, and free-flow time to the simulated horizon; it is an approximation, not an exact remaining clearance time. The oracle uses the realized capacities with zero capacity-scenario

spread, so its difference from conditioning also includes this change in dispersion.

Observation, candidate, and scenario substreams are derived independently at each control update, preventing state-dependent candidate sampling from shifting the scenario stream. The primary scenario coupling is explicit. For the same standard-normal draw ε_j^s and the same compliance draw in scenario s,

$$m^s_{\mathrm{uncond},j} = \mathrm{clip}\big(\exp[-\sigma^2/2 + \sigma\varepsilon^s_j]\big)\,, \tag{12}$$

$$m^s_{\mathrm{cond},j} = \mathrm{clip}\big(\widehat{m}_j \exp[-\sigma^2/2 + \sigma\varepsilon^s_j]\big)\,. \tag{13}$$

Thus the center changes while the pre-clipping log-spread parameter, clipping rule, and random variates remain fixed. A secondary contracted-spread policy uses $\max(0.05, 0.45\sigma)$ solely to quantify that separate design choice.

The base benchmark has 5000 people, four zones, and three corridors. Controllers update every 120 s with default $\Delta t = 10$ s. The archived primary validation uses seeds 100000–100199 and 5000 paired-bootstrap resamples. Environment and policy seeds are separate; matched policies share each run's realized capacities and compliance. The frozen CSV tables are the numerical source for this revision. Diagnostic protocols use 50 seeds per uncertainty–compliance cell, 30 per time step, 50 per estimator regime, 50 per network family, and 30 per sensor regime. The time-step sweep holds the control interval, rollout horizon, and estimator decay in physical time fixed. Structural transfer covers the base network, symmetric, path-bottleneck, exit-bottleneck, and expanded 5-zone/4-corridor configurations at $\sigma = 0.30$ and $\mu = 0.68$. Controller-facing estimate faults include multiplicative log-noise 0.15, 60-s delay, 20% dropout, and $+10\%$ bias. These modify the already-computed estimates passed to the controller; queue states, raw throughput and saturation classification remain ideal. They are evaluated in the default $(0.22, 0.68)$ and high-information $(0.45, 0.85)$ (σ, μ) regimes.

A. Independent Same-State Diagnostics

The additional diagnostic protocol uses 30 previously unused environmental seeds, 110000–110029, with policy seed equal to environmental seed plus 1000003. Both primary mean policies collect pre-decision states at 120, 360, and 600 s, giving at most 180 snapshots. A state already cleared at a sampling time is recorded as skipped and is not replaced. Saved states retain all queues, transit, estimator histories, previous recommendations, accumulated person time, and pre-decision policy/RNG state. Trace collection reuses the actual decision evaluation without consuming additional random draws.

At each snapshot, the selected and reactive recommendations are compared under the same realized capacities and compliance. One continuation holds each recommendation fixed to clearance. The other applies it for 120 s and then uses reactive guidance, recomputed every 120 s on that branch's own state. Outcomes are residual times from the snapshot; the absolute 3600-s limit is retained and incomplete branches are flagged. Equivalent recommendations count as reactive choices. We log full-objective gaps separately from clearance-component gaps, and compare the latter with realized fixed-action clearance ordering. The common-reactive continuation measures consequences of the current action; it is not a comparison of complete RH and reactive policies. Intervals use 5000 seed-cluster bootstrap resamples, retaining dependence among phases and trajectories from each environment.

TABLE I
PRIMARY PAIRED VALIDATION ($n = 200$)

Policy	Mean (s)	P90 (s)	Mean peak (p/m^2)
Static nearest	981.30	1200.0	5.343
Reactive balance	708.80	820.0	2.849
Unconditioned mean	721.45	830.0	2.799
Conditioned mean	705.80	801.0	2.644
Oracle-capacity mean	696.90	791.0	2.587

Runtime uses the first five diagnostic seeds and the same eligible snapshots. Reactive, unconditioned mean, and conditioned mean decisions are measured serially, with one BLAS thread, five warmups and 30 timed repetitions per policy and snapshot. Every repetition resets state, policy and RNG; cloning, construction and file I/O are excluded. Cyclic measurement order reduces order effects. We report median and 95th-percentile decision times with hardware and software metadata.

V. RESULTS

A. Primary Validation and Observability

Reactive balancing improved substantially over static routing (-272.50 s, 95% interval $[-291.10, -254.90]$), whereas unconditioned lookahead differed from reactive control by +12.65 s ($[+5.95, +19.60]$). The center-only comparison averaged 705.80 s versus 721.45 s, giving $\Delta T_{\mathrm{sense}} = -15.65$ s (-2.17%; $[-20.75, -10.80]$). The difference between marginal P90 values was -29.00 s ($[-60.00, -10.00]$); win/tie/loss was 45.0%/46.0%/9.0%, and the worst observed paired harm was 70.0 s. The direct conditioned-minus-reactive effect was -3.00 s ($[-7.50, +1.35]$), about 0.42% of the reactive mean. Its interval includes zero. The main 2.17% benefit therefore describes improvement over unconditioned RH, not an established advantage over reactive guidance. The oracle was 8.90 s faster than conditioned control (oracle-minus-conditioned interval $[-11.85, -6.25]$ s); zero spread, finite candidates, and finite lookahead limit its interpretation.

The contracted-spread diagnostic changed clearance by -5.00 s relative to the equal-spread conditioned controller ($[-7.50, -2.75]$). This separates any benefit from narrowing the scenario distribution from the primary value of moving its center.

Figure 2(d) reports the information mechanism directly. In the default regime, combined capacity-multiplier MAE changed from 0.163 at the first recorded step to 0.122 at 240 s, when 28.9% of links had been observed under saturation. At clearance, mean MAE was 0.106 and mean observed fraction was 37.2%. At clearance, path MAE was 0.1300 with 22.17%

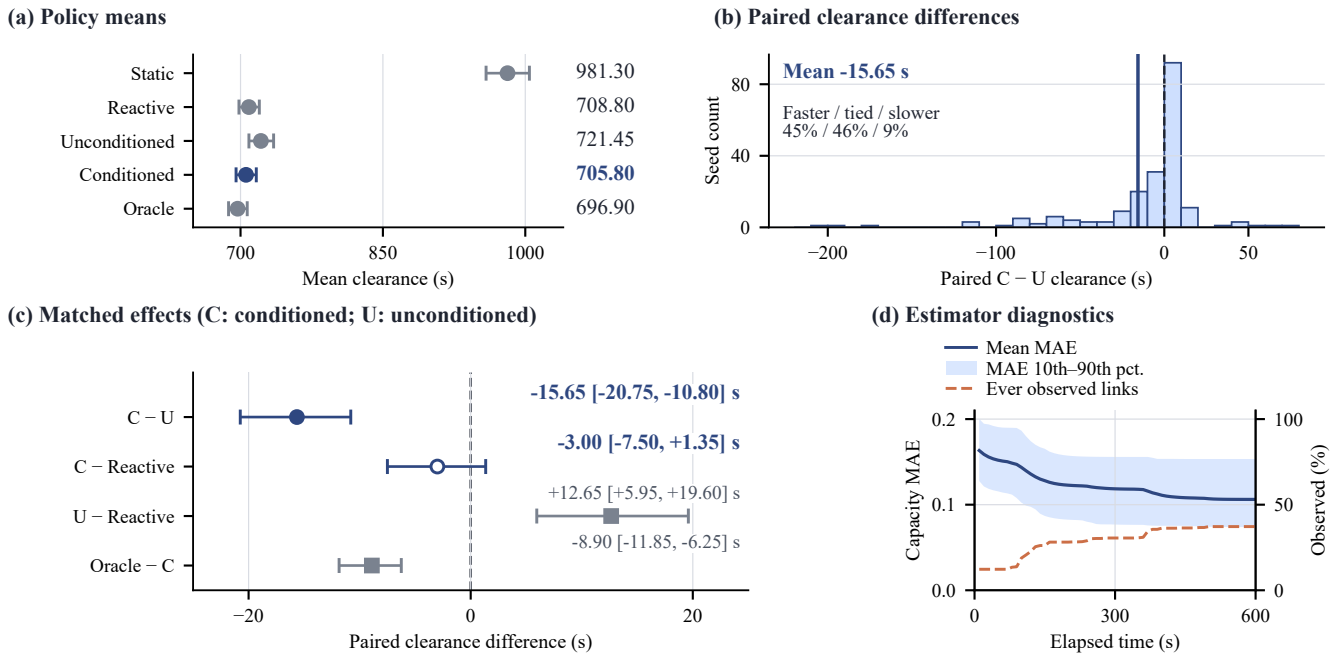


Fig. 2. Primary evidence. (a) Discrete policy means with marginal 95% bootstrap intervals; runs are paired by seed. (b) Paired conditioned-minus-unconditioned clearance differences. (c) Paired mean effects and 95% bootstrap intervals; filled markers exclude zero and hollow markers include zero. Negative differences favor the first policy. (d) Mean capacity-multiplier MAE and mean fraction of links ever observed saturated. Blue shading is the 10th–90th percentile range of run-level MAE across 50 seeds at each time, a descriptive range rather than a confidence band.

of paths ever observed saturated; exit MAE was 0.0107 with 97.33% coverage. This pattern is consistent with informative exit bottlenecks, but does not isolate their causal contribution without path-only/exit-only ablations. These are benchmark diagnostics rather than field-sensor accuracy claims.

B. Operating Regions and Robustness

The map in Fig. 1(c) shows that information becomes actionable jointly through uncertainty and compliance. At $\sigma = 0.45$, conditioned-minus-unconditioned effects across $\mu \in \{0.35, 0.50, 0.68, 0.85\}$ were -8.6, -16.8, -41.2, and -40.0 s, respectively. At negligible uncertainty, effects were correspondingly small or unresolved. Because each map cell is diagnostic and not multiplicity-adjusted, the map supports regional patterns rather than cell-level guarantees.

The matched effect was -17.7, -30.2, and -29.1 s at time steps of 10, 5, and 2.5 s, respectively. Across the five network families, 5 of 5 effects had intervals entirely below zero; point estimates ranged from -29.8 to -2.6 s. The network study varies structural bottlenecks and size, but remains synthetic.

Under clean controller-facing estimates, the effect was -14.7 s in the default regime and -63.7 s in the high-information regime. Across the four faults, estimates ranged from -20.0 to -13.3 s and from -64.0 to -46.3 s, with every interval below zero. At high information, delay and dropout attenuated the clean point estimate, whereas noise and bias were nonmonotone. These checks test errors in the estimates delivered to control, not the complete raw-sensing and saturation-classification pipeline.

C. Candidate Selection and Computation

Of 180 planned snapshots, 180 were eligible; 720 of 720 counterfactual branches completed. Unconditioned control selected a recommendation equivalent to reactive in 0.0% of snapshots. Its fixed-action clearance ranking reversed in 28.9% (95% seed-bootstrap interval [20.0, 37.8]%). With the common reactive continuation, the selected-minus-reactive action effect was -0.67 s ([-5.44, +4.44] s). Conditioned control selected a recommendation equivalent to reactive in 4.4% of snapshots. Its fixed-action clearance ranking reversed in 14.4% (95% seed-bootstrap interval [8.9, 20.0]%). With the common reactive continuation, the selected-minus-reactive action effect was -3.56 s ([-7.56, +0.11] s). Both action-effect intervals include zero. Reversal frequencies use all completed decisions as denominator; predicted or actual ties are not reversals and are separately tabulated. The diagnostics demonstrate that forecast ranking need not persist under realized capacity; they do not apportion the contributions of capacity mismatch, terminal approximation, finite scenarios and objective choice. The two reversal rates describe different on-policy state sets; their difference is not an isolated effect of conditioning.

On AMD64 Family 26 Model 104 Stepping 0, AuthenticAMD, Python 3.14.6 and NumPy 2.5.3, the benchmark recorded 2700 timed decisions over 30 shared snapshots. Both predictive controllers had similar latency, with overall p95 at most 4.59 ms; reactive control was substantially cheaper. These measurements exclude state copying and I/O and provide no hard real-time or deployment guarantee.

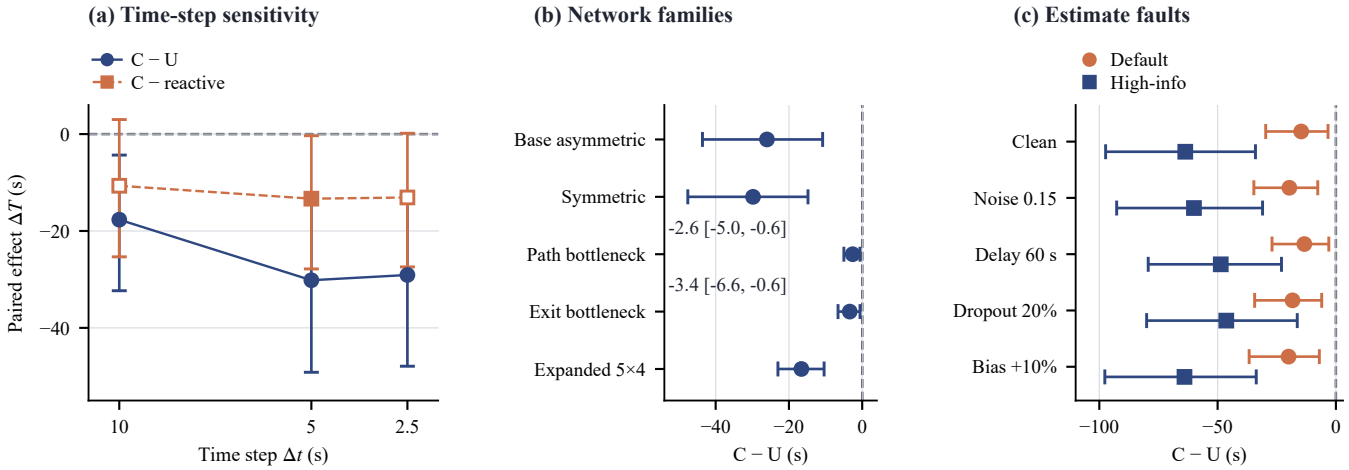


Fig. 3. Robustness and estimate-fault diagnostics (C: conditioned; U: unconditioned). (a) Effects at the three tested time steps. (b) C-minus-U effects across five synthetic network families ($\sigma = 0.30$, $\mu = 0.68$). (c) Controller-facing estimate faults in the default and high-information regimes. Intervals are paired-bootstrap 95% intervals; filled markers exclude zero and hollow markers include zero. Negative differences favor conditioning.

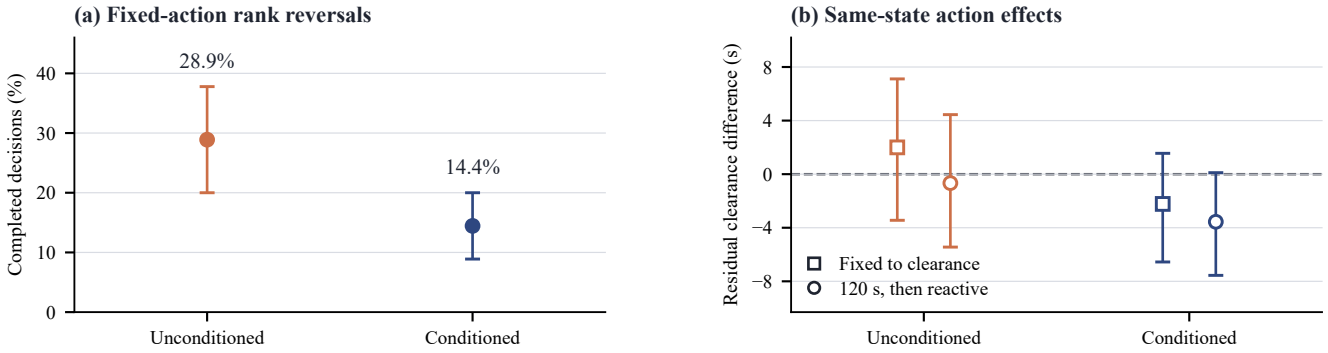


Fig. 4. Same-state diagnostics. (a) Strict predicted-versus-realized fixed-action clearance-rank reversals among all 90 completed decisions per collector; ties are not reversals. Collectors visit different state sets, so their rate difference is descriptive. (b) Selected-minus-reactive residual clearance effects: squares hold actions fixed; circles apply actions for 120 s then use reactive guidance. Actions are paired within each snapshot. Intervals resample environmental seeds (95%); hollow effect markers include zero, and negative effects favor the selected action.

TABLE II
DECISION RUNTIME ON SHARED DIAGNOSTIC STATES

Controller	Median (ms)	p95 (ms)
Reactive	0.021	0.039
Unconditioned mean	4.050	4.590
Conditioned mean	4.042	4.515

VI. DISCUSSION

The center-only design supports an internal causal interpretation within the implemented benchmark: the programmed intervention changes the location of the capacity predictions, not their log-spread parameter, clipping rule, candidate architecture, objective, or scenario random draws. The subsequent divergence in states and selected actions is an endogenous consequence of that intervention rather than an additional experimental change. The operating map shows that the benefit depends on uncertainty and compliance in the tested configurations. The observability summaries are consistent with information about bottlenecks being useful; identifying which links drive the effect would require additional ablations.

External validity remains the principal limitation. The simulator omits continuous pedestrian interaction, group behavior, detailed geometry, counterflow, collision avoidance, pressure, smoke, and injury risk. Saturated throughput is deliberately informative, all queue states are available, and the network families are designed rather than empirically calibrated. Real systems would require saturation-classification error, aggregation, privacy, latency, missing-data, change detection, venue calibration, and professional safety review [18]. At high σ, clipping shifts the realized environmental mean; reported effects use that clipped distribution. A shifted center near a bound can also change post-clipping moments, although both controllers

apply the same transform. The EWMA already discounts older observations. Tracking an abrupt capacity change would nevertheless depend on obtaining new saturated observations and on estimation and control update times; incident tracking and change detection require separate validation.

The inclusion of a reactive candidate does not guarantee that RH clears faster than the reactive policy. A candidate is held constant for 300 s in prediction, whereas the executed reactive policy recomputes its recommendation every 120 s. Moreover, minimizing Eq. (11) need not minimize realized final clearance: the objective also values person time, density and switching. Nominal capacity centers, finite scenario samples and the terminal approximation can alter the ranking. These explain the absence of a guarantee. The nonzero fixed-action rank-reversal rates document disagreement between the clearance surrogate and realized fixed-action ordering. However, the mean effects in both counterfactual continuations have intervals that include zero. The diagnostics neither establish a selected-action clearance advantage nor apportion the full-policy difference among capacity centers, finite scenario sampling, terminal approximation and objective trade-offs.

The primary benefit is modest: the final clearance instant is earlier by about 16 s in a roughly 12-min process, not a 16-s saving for every person. The paired median is zero and some runs are slower. For an existing RH system with usable throughput data, the within-controller comparison is relevant to adding capacity estimation. For a system currently using reactive guidance, the direct comparison is the relevant adoption evidence and remains unresolved here. Runtime quantifies computation on the measured machine; instrumentation, calibration, maintenance and deployment costs are unmeasured. Neither these timings nor the queue-density proxy establish an operational safety gain.

VII. Conclusion

Updating persistent-capacity forecast centers improved mean clearance within the specified RH benchmark by 15.65 s (2.17%). The direct advantage over reactive guidance remains unresolved. The matched comparison, observability summaries and independent decision diagnostics separate a within-controller information benefit from a general claim for predictive guidance. The conclusions concern synthetic fixed-capacity networks with ideal state observations; raw-sensor faults, abrupt incidents and venue-calibrated performance remain unvalidated.

AI Assistance Disclosure

Generative AI tools assisted manuscript revision and research-code development. Responsibility for the final manuscript, code, and interpretation remains with the authors.

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